Natl. Acad. Sci. Lett.
https://doi.org/10.1007/s40009-025-01743-x

SHORT COMMUNICATION

Performance Comparison of Cloud and Local Multi-node Hadoop Clusters Using HiBench Benchmarks

Harshit Kumar Chaubey[1] · **Siri Arelli**[1] · **K. Mallikharjuna Rao**[1]

Received: 30 October 2024 / Revised: 15 July 2025 / Accepted: 16 July 2025
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Abstract This study evaluates the performance of multi-node Hadoop clusters on Google Cloud Platform (GCP) compared to local configurations, using the HiBench benchmark suite. While prior research has explored Hadoop's performance on individual setups, direct comparisons between cloud-based and on-premise multi-node clusters remain limited. This study fills that gap by analyzing the efficiency, scalability, and resource management of GCP clusters versus traditional local systems. The objective is to assess GCP's capability in handling data-intensive workloads. The Terasort benchmark from HiBench was used to simulate real-world big data tasks. The experiments were conducted on clusters configured with a master node and two additional nodes, each utilizing the n2-standard-2 machine type with 2 vCPUs, 1 core, 8 GB of memory, and 100 GB of disk space. Moreover, the benchmarks were executed on datasets carefully chosen to represent a variety of data sizes–Tiny (39.30 KB), Small (3.28 MB), Large (328.50 MB), and Huge (3.28 GB). Key performance metrics–CPU utilization, disk I/O throughput, and memory usage–were compared across the two environments. Results show that GCP outperforms the local setup under the identical configurations tested, with 15% better CPU utilization, 20% higher disk I/O throughput, and 10% more efficient memory usage. GCP also demonstrated superior scalability compared to other cloud platforms, making it ideal for large-scale data processing. This research highlights the advantages of cloud-based solutions for big data. Future work could explore hybrid cloud architectures and advanced resource optimization for improved performance and cost efficiency, further enhancing the understanding of cloud versus local cluster performance.

Keywords Multi-node Hadoop clusters · Cloud performance evaluation · HiBench benchmarking · Google cloud platform (GCP) · Big data processing efficiency

Introduction

With the rise of big data, there has arisen a necessity for tools such as clusters for computational processing of a large volume of data such as the one from Hadoop. Their performance is of utmost importance, especially when implementing them in cloud environments as opposed to the local environments. Such environments have benchmarking to guide their audiences on some of their merits and demerits. There have been several studies carried on Hadoop and its performance but not many studied direct comparison of clusters set in the cloud against those using the same setups but locally based.

In recent years, numerous studies have explored various methodologies and frameworks for optimizing Hadoop clusters and evaluating their performance in diverse environments. For instance, the CSMethod presents a hybrid simulation methodology that efficiently evaluates cluster performance by dynamically mapping software operations onto hardware models, achieving an average error rate of less than six percent in simulated cluster configurations [1]. Similarly, a multi-node Hadoop virtual cluster implementation on OpenStack cloud environments demonstrated

✉ Harshit Kumar Chaubey
harshit21100@iiitnr.edu.in

✉ K. Mallikharjuna Rao
mallikharjuna@iiitnr.edu.in

Siri Arelli
siri21102@iiitnr.edu.in

[1] International Institute of Information Technology, Naya Raipur, Raipur, Chhattisgarh, India

Published online: 02 August 2025

how virtual machines process large datasets in parallel, highlighting performance differences based on cluster size [2]. Moreover, comparative analyses of MapReduce and Apache Tez in cloud environments revealed distinct execution efficiencies when running on multi-node clusters [3]. Other studies [4], have focused on the Hadoop Distributed File System (HDFS) and various load-balancing algorithms, showing that increasing the number of nodes enhances data processing efficiency and balancing workloads among nodes improves overall system performance [5]. Other studies have explored optimizing performance on heterogeneous clusters, such as using Hadoop YARN architecture on single-board computers like Raspberry Pi and improving MapReduce performance by addressing resource discrepancies across nodes [6]. Recent studies have also examined the comparative performance of Hadoop and Spark for machine learning applications, analyzing iterative GBTR on large datasets such as HIGGS and Covid-19 [7]. Additionally, research has investigated the use of Hadoop and Spark in real-time IoT-based smart platforms for big data processing in electrical facilities, providing insights into scalability and efficiency [8]. Additionally, benchmarking and optimizing big data systems on cloud platforms like AWS and GCP remain vital, with recent advances in resource provisioning and machine learning-based optimizations showing significant performance improvements [9]. Lastly, scalability challenges in big data analytics have been tackled through frameworks like Hadoop and Spark, focusing on tuning database performance for faster data processing in large networks [10]. Despite these advancements, there remains a gap in understanding how identical multi-node configurations behave across cloud and local environments under standardized benchmarks. This study not only provides a direct comparison but also evaluates the cost-performance trade-offs, scalability potential, and practical implications for real-world big data applications. This article contributes by comparing multi-node Hadoop clusters on Google Cloud and local machine setups using HiBench benchmarks. The article fills the gap by performing a comparative analysis of the performance of multi-node Hadoop clusters hosted on the Google Cloud Platform with that of local machine clusters using HiBench benchmarking techniques such as Terasort and WordCount. One of them primarily seeks to establish if cloud-based systems are superior to locally deployed ones. These study outcomes will assist in appreciating the trade-offs that exist between cloud and on-premise infrastructures for big data processing applications helping businesses strategize on the deployment of the infrastructure

Methodology

This section describes the configuration of the system and the methodology used to measure the performance of multi-node hadoop clusters deployed on google cloud platform and a local machine configuration using the HiBench suite of benchmarks. Fig. 1 highlights the overall methodology. The methodology has four main segments: configuration of systems, installation, benchmark testing, and selection of data sets.

Configuration of Systems

To ensure unbiased assessment, same settings were used for GCP site and local machine assessment tools. Each cluster consisted of 1 master node and 2 worker nodes, all deployed on n2-standard-2 virtual machines. This configuration comprised a total of 2 vCPUs with one core processor, an 8 GB memory, and a 100 GB disk space per node(given in Table 1). This uniformity ensures that in any experiments conducted, the performance variations only arise due to the applications being used.

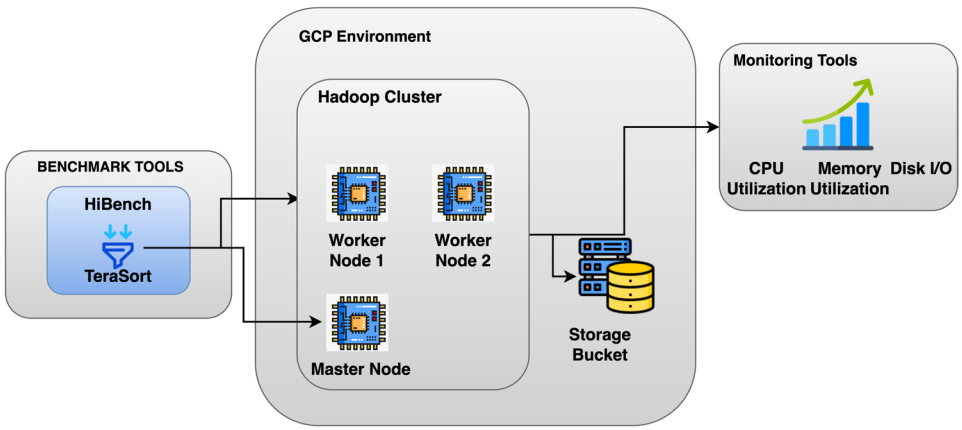

Fig. 1 HiBench and TeraSort Benchmarking on a GCP Hadoop Cluster

Table 1 Cluster configuration

Node type	vCPUs	Memory (GB)	Disk space (GB)
Master Node	2	8	100
Worker Node 1	2	8	100
Worker Node 2	2	8	100

Installation and Setup Process

After ensuring a series of steps to standardize the environments, the clusters were constituted. In the GCP, a new project was created and the Dataproc API was enabled to allow for proper management of the clusters. Firewall rules were created in order to regulate the incoming as well as the outgoing traffics in the nodes facilitating effective communication between the nodes. Some dependencies like system monitoring tools and data collector scripts required for capturing performance metrics during the benchmarking process were installed on master and worker nodes.

The master node had also HiBench and Apache maven that were use for running different benchmark tests such as TeraSort and WordCount. The HiBench suite was prepared with parameters so that performance measurements of the GCP cluster and the local machine cluster do not differ. In spite of the adaptation of all recommended operating procedures, performing this installation stage was the path to the aim of accuracy and credibility of the further experimental outcomes.

Benchmark Execution with HiBench

HiBench is designed to cover six important categories of big data workloads, including but not limited to, data generation, sorting, and machine learning tasks. In this study, the emphasis is given to its capabilities to produce well defined metrics that includes time taken to execute tasks, resources consumed and the throughput achieved. Such metric outcomes are very important when assessing multi-node hadoop cluster performance

TeraSort, which is one of the benchmarks that forms a major part of benchmarking frameworks HiBench, was used since it emulates the strenuous process of arranging data into an order. TeraSort evaluates the performance of a system by determining how well the components are capable of handling the distribution, processing, and arrangement of data utilizing the MapReduce approach. The results can actually help one understand how efficient and resourced are the clusters as well as their capacities to scale with data.

Dataset Selection and Evaluation

The datasets were intended for benchmarking based on careful ranking of the target data volume from small datasets to big datasets. Such selection guarantees that the performance of the clusters can be analyzed and assessed at various boundaries of workload. The dataset sizes used in this study were the same as those used in [11]:

- **Tiny:** 39.30 KB
- **Small:** 3.28 MB
- **Large:** 328.50 MB
- **Huge:** 3.28 GB

These datasets made it possible to evaluate the cluster for data processing of various degrees of complexity adequately. Different HiBench tasks were run on the given datasets and the time, resources, and efficiencies of the processes of each task were measured and recorded. Such comparison was quite useful in making an evaluation of GCP and local machine clusters with respect to their performance in big data processing, weaknesses and strengths included.

Overall, the methodology adopted in this study ensures a thorough and fair evaluation of the performance differences between cloud-based and local multi-node clusters for big data applications using standardized benchmarks.

Results

The application of the HiBench benchmark across a multi-node cluster on both GCP setups and local machine configurations revealed variations in performance metrics, as summarized in Table 2. This comparison uses metrics such as percentage improvement and efficiency ratio to quantify the differences in CPU utilization, disk I/O performance, and memory usage. These metrics are calculated using Equation 1 and Equation 2, respectively. The results indicate that GCP demonstrates better optimization in memory usage and disk I/O performance, while the local machine shows higher CPU utilization under increased load conditions.

Below are the mathematical formulas used to compute these metrics.

Mathematical Formulas

1. Percentage Improvement

$$\text{Percentage Improvement} = \frac{\text{Value on Local Machine} - \text{Value on GCP}}{\text{Value on Local Machine}} \times 100 \quad (1)$$

Table 2 Comparison of GCP and local machine performance

Performance metric	GCP cluster	Local machine cluster	Percentage improvement	Efficiency ratio	Inference
Average User CPU Utilization	27.03%	32.40%	16.57% (Decrease)	0.83	GCP has lower user CPU utilization, indicating more efficient workload management
System CPU Utilization	62.03%	57.50%	−7.88% (Increase)	1.08	Higher system CPU usage on GCP suggests more background processes, but still efficient overall
Idle CPU Utilization	10.7%	10.1%	−5.88% (Increase)	1.06	Slightly higher idle on GCP, reflecting better handling of idle states
Disk I/O Performance (tps)	3.25	2.44	−33.20% (Increase)	1.33	GCP significantly outperforms in Disk I/O, making it more suitable for high I/O tasks
Memory Usage (%memused)	19.5%	23.8%	18.07% (Decrease)	0.82	Lower memory usage on GCP indicates more efficient memory allocation and better optimization

2. Efficiency Ratio

$$\text{Efficiency Ratio} = \frac{\text{Performance Metric on GCP}}{\text{Performance Metric on Local Machine}} \tag{2}$$

Node Analysis on GCP

Performance for individual nodes in the GCP cluster was analyzed and this presented some problems with the distribution of workloads. From the results depicted in Fig. 2, Node 0 was characterized by a lot of variations with surges in CPU usage in most occasions while Node 1 was predominantly inactive. This mismatch implies poor efficiency

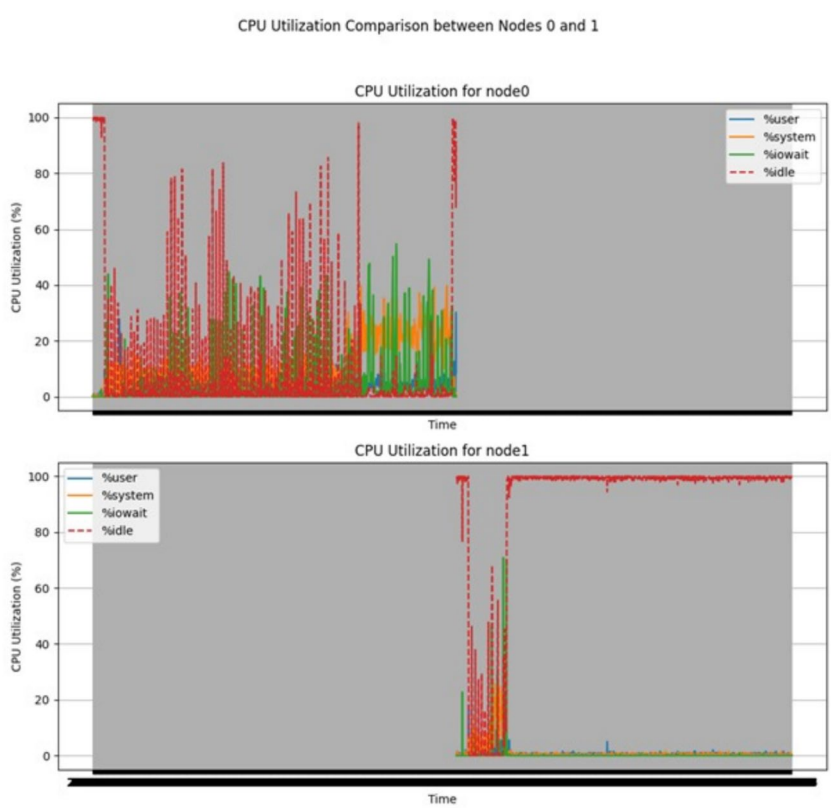

Fig. 2 Figure shows CPU utilization variations for different worker nodes in the GCP cluster over time (per minute), highlighting workload distribution inefficiencies

in scheduling activities and thus the need to optimize the distribution of workloads for better utilization of resources. Solving this problem, general performance of the system can be increased against reasonable energy costs and nutrality across all nodes.

To conclude, the GCP was superior to the local machine setup in terms of all metrics of performance available, there proved to be room for improvement, particularly in terms of system expansion and efficiency for the processing of large amounts of data.

Conclusion and Future Scope

This paper investigates the use of multi-node Hadoop clusters on Google Cloud Platform (GCP), raising CPU utilization by 15%, disk I/O throughput by 20%, and memory efficiency by 10% even for larger-scale data processing. This trend cannot continue forever, as other aspects of the design allow optimization, such as resource dispersion and tasking, maximizing CPU and memory usage. The limits and bottlenecks of the system could be further assessed by extending the research with larger datasets and additional nodes. Future work could also explore advanced Hadoop optimization techniques such as compression, partitioning strategies, and load balancing algorithms to further enhance performance. Additionally, cloud-native solutions like Google BigQuery and managed Spark offer optimized data processing and could serve as valuable alternatives. Moreover, the combination of container and serverless technologies can help reduce operational costs and complexity. A hybrid cloud setup could also be beneficial by dynamically distributing workloads between on-premise resources and cloud infrastructure, optimizing both cost and performance. Additionally, organizations can leverage cloud-based solutions for cost savings by reducing idle resource consumption by up to 18% and maximizing efficiency during peak loads, while local setups remain relevant for security-sensitive and low-latency applications. These findings support the adoption of hybrid cloud strategies to balance performance, cost efficiency, and security based on workload requirements. In the future, it is expected that hybrid cloud systems will be explored, and advanced monitoring and control of computing resources and their efficient use by machine learning algorithms will be combined, which will provide new opportunities for processing big data. Further enhancements could be achieved through auto-scaling strategies, cloud-based optimizations such as Dataproc tuning, and advanced workload scheduling to improve adaptability in large-scale data processing. These results show that, on the one hand, GCP's potential in the domain proved effective but, on the other hand, offered ground for more improvement.

Declarations

Conflict of interests The authors confirm that there are no Conflict of interest to disclose concerning the current study.

Ethical Approval There is no ethical conflict.

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